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Tracking knowledge proficiency of students with calibrated Q-matrix

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ABSTRACT

With the emergence of intelligent educational systems, numerous research works are dedicated to *Knowledge Tracing* (KT), which refers to the issue of diagnosing students' changing knowledge proficiency in exercises. Recent developments in KT have yielded immense success on this task and they mainly use sophisticated and flexible deep neural network-based models to fully exploit the interaction information between students and response logs. However, these models either ignore the significance of Q-matrix associated exercises with knowledge concepts (KCs) or fail to avoid the subjective tendency of experts within the Q-matrix. To tackle these problems, in this paper, we devise a novel *Calibrated Q-matrix-based Knowledge Tracing* (CQKT) framework to track knowledge proficiency of students dynamically in KT. To be specific, for the original Q-matrix, we primarily strive to capture the high-order connectivity between exercises and KCs to obtain potential KCs of each exercise by utilizing graph convolution network. Then, three Q-matrix calibration methods based on a pairwise Bayesian treatment equipped with potential KCs are adopted to refine and calibrate the raw Q-matrix so that the subjective tendency of the Q-matrix defined by domain experts can be weakened. After that, the embedding of each exercise aggregated the calibrated Q-matrix with historical student interactions is injected into the Long Short-Term Memory (LSTM) network to trace students' knowledge states. Extensive experiments are conducted on three real-world benchmark datasets and the results show the superiority of CQKT. In particular, we demonstrate its practicability via applying it to three fundamental educational tasks, including score prediction, knowledge state estimation, and diagnosis result visualization.

1. Introduction

Recent advances of the technologies for intelligent tutoring systems create an open as well as convenient environment for students to learn and do exercises individually, by providing a tremendous amount of learning resources to them (e.g. Chen, Liu, Huang, Le, Chen, Wu, Su, & Hu, 2017; e.g. Wang, Ma, Zhao, Li, & He, 2021). Although students' attention is grabbed by those various educational platforms provisionally, recent studies indicate that students usually tend to lack of enthusiasm and show high dropout rate in practice (e.g. Huang et al., 2020). To address these concerns, personalized learning is an effective way that involves the concise extraction of learning resources desired by students rather than the imposition of extra burdens on them (e.g. Zhou, Huang, Hu, Zhu, & Tang, 2018).

Among them, tracing students' knowledge states (KS) lies at the core of personalized learning, with the goal of estimating how much they master on every single knowledge concept (e.g. Liu et al., 2018; Yeung and Yeung, 2018). Generally, students usually first select some items from a set of exercises (e.g., e_1, e_2, e_3) to practice and leave their response logs (e.g., right or wrong) as shown in Fig. 1. Then, the purpose of KT is to infer their latent knowledge states on each KC through their past attempts on exercises and exercise-KC interactions, where the cross-mapping between exercises and KCs is named as Q-matrix whose row and column indicate exercise and KC respectively. Actually, these estimated knowledge states of students are remarkably useful in numerous scenarios, such as target knowledge training (e.g. Liu et al., 2021) and personalized exercise recommendation (e.g. Wang et al., 2020a). In terms of educational measurement, cognitive diagnosis models (CDMs) usually offer a psychometric framework with the help of Q-matrix to assess students' overall ability by a latent trait value (e.g. Huo, Wong,

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Ni, Chao, & Zhang, 2020a) or a binary skill mastery vector (Jimmy, 2009). Comparatively, depending on treating such a diagnostic task as a data-mining problem of score prediction to trace students' KS in an indirect manner, deep learning techniques encode interactions between students and exercises in a series of sequential input vectors when making predictions (e.g. Piech, Spencer, Huang, Ganguli, Sahami, Guibas, & Jascha, 2015). Unfortunately, most existing methods designed for KT task either do not take the relationships between exercises and KCs into account (i.e., Q-matrix) or are incapable of eliminating the subjective tendency of Q-matrix specification from subject matter experts.

As the matter of fact, it is of great necessity to leverage Q-matrix for diagnosing and tracing KS of students, since Q-matrix is the source of information about the exercise knowledge concept (e.g. Wang, Cai, & Tu, 2020b). In its basic form, the Q-matrix is binary, with 1′s for association representing that which KC is covered by an exercise, and similarly, 0′s for non-association (e.g. Jimmy and Chiu, 2016) (an example is shown in Fig. 2). However, most of the existing efforts devoted to KT task are not able to explore more implicit semantic information as further as possible. On one hand, deep learning methods ranging from Deep Knowledge Tracing (DKT) (e.g. Piech et al., 2015) to Dynamic Key-Value Memory Network (DKVMN) (e.g. Zhang, Shi, & King, 2017) almost abandon the traditional Q-matrix. On the other hand, cognitive diagnosis models with Q-matrix generally ignore structural and inherent information. Considering that Q-matrix is proved to be directly useful, our work aims to exploit the association between exercises and KCs with Q-matrix and apply it to track the knowledge proficiency of students.

Although Q-matrix contains implicit and adequate semantic information, in practice, Q-matrix is typically annotated by educational experts, which tends to be subjective and likely results in some errors (e.g. Wang et al., 2020b). What is worse, Q-matrix with subjective bias and errors has negative impact on the estimation of the model parameters (e.g. Jimmy, 2009; Rupp & Templin, 2008). Given the inevitable errors and subjective bias in manually-labeled Q-matrix, existing researches attempted to conquer this problem from the perspective of data-mining (e.g. Huo, Wong, Ni, Chao, & Zhang, 2020b) and cognitive diagnosis (e.g. Huang et al., 2020; Wang et al., 2020b, 2020c; Xu & Shang, 2018). In data-mining community, Q-matrix estimation method (e.g. Wang et al., 2020c) is a topic of high practical values but has been relatively little explored in this research community. Huo et al. (2020b) has derived a contextualized Q-matrix to weaken the subjective bias from human experts but their method fails to reveal the inherent relationships between exercises and KCs. Despite of Q-matrix estimation approaches proposed in the domain of cognitive diagnosis such as (e.g. Wang et al., 2020c) and (e.g. Xu & Shang, 2018), these methods are highly relying on model-data fit of specific cognitive diagnosis models. In this work, we primarily pay attention to tackling the following two challenges: How to exploit the semantic information in Q-matrix as fully as possible in an effective and reasonable manner to improve the performance of KT model? How do we eliminate the subjective tendency in Q-matrix to a great degree without depending on specific cognitive diagnosis models?

To address these challenges, in this paper, we keep abreast of the ongoing developments of knowledge tracing techniques (e.g. Piech et al., 2015; Huo et al., 2020b; Zhang et al., 2017) and pertinent cognitive models (e.g. Embretson & Reise, 2013; Jimmy, 2009; e.g. Wang et al., 2020b), and propose a *Calibrated Q-matrix-based Knowledge Tracing* (CQKT) framework to track the mastery level of students and further develop three methods incorporating Graph Convolution Network (GCN) (e.g. He et al., 2020) with a pairwise Bayesian treatment to eliminate the subjective tendency of experts within the Q-matrix. Concretely, we first attempt to exploit the high-order connectivity between exercises and KCs within an exercise-KC interaction graph (i.e. Q-matrix) via GCN to obtain a list of potential KCs of each exercise. Then, three Q-matrix calibration approaches are designed by leveraging the list of potential KCs of each exercise to eliminate the subjective tendency of the Q-matrix. Furthermore, the embedding of each exercise represented by the calibrated Q-matrix along with historical student interactions as input is sent to the LSTM to track students' knowledge proficiency on each KC. Finally, extensive experimental studies on three real-world benchmark datasets show that the proposed framework enables to trace and diagnosis actual KS of students effectively. Meanwhile, the practicability of this framework is also verified by three fundamental diagnostic tasks involved in score prediction, knowledge state estimation, and diagnosis result visualization. In summary, our contributions are three folds:

(1) A novel *Calibrated Q-matrix-based Knowledge Tracing* (CQKT) framework based on GCN is developed. To the best of our knowledge, this is the first attempt to calibrate Q-matrix by using GCN theory to avoid the subjective tendency existed in Q-matrix, which has been heavily studied in the area of educational measurement for many years.
(2) Three Q-matrix calibration methods via a pairwise partial order between KCs are presented for improve knowledge tracing. In effect, with the help of the calibrated Q-matrix, most deep knowledge tracing models can be interpretable whereas hidden vectors in those conventional deep models fail to match with every KC.
(3) In practice, extensive experimental results on three real-world datasets show that our framework enables to track the knowledge proficiency of students effectively and have impressive practicability on three fundamental diagnostic tasks, such as student score prediction, knowledge proficiency estimation and diagnosis result visualization.

The rest of this paper is organized as follows: we briefly review the related work in Section 2. Some preliminaries and important notations are specified in Section 3. In Section 4, the proposed CQKT framework is presented and discussed in detail. We conduct extensive experiments on three real-world datasets and comprehensively explore these results in Section 5. Section 6 concludes the paper and identifies future work which is likely to be carried out on the basis of the new findings exhibited in this paper.

2. Related work

Our proposed framework is pertinent to three broad strands of the research literature: first, we look into several widely used and classic cognitive diagnosis models to present their effectiveness of modeling student's knowledge states and their limitations to Q-matrix; then, we

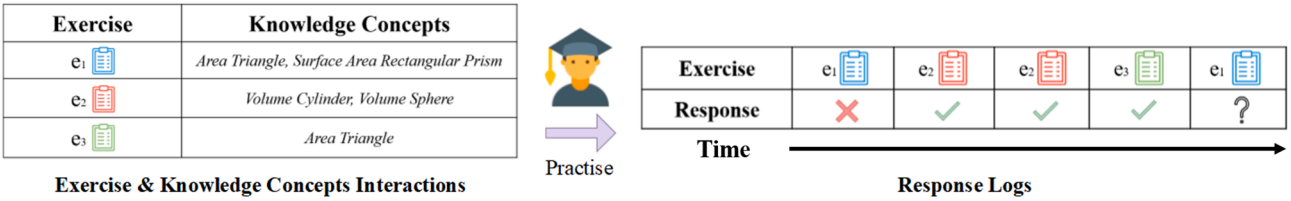


Fig. 1. A showcase of KT task for a student on mathematical exercises. The student chooses some exercises from different knowledge points for practice and leave the response logs. Now the student is going to answer exercise e_3.

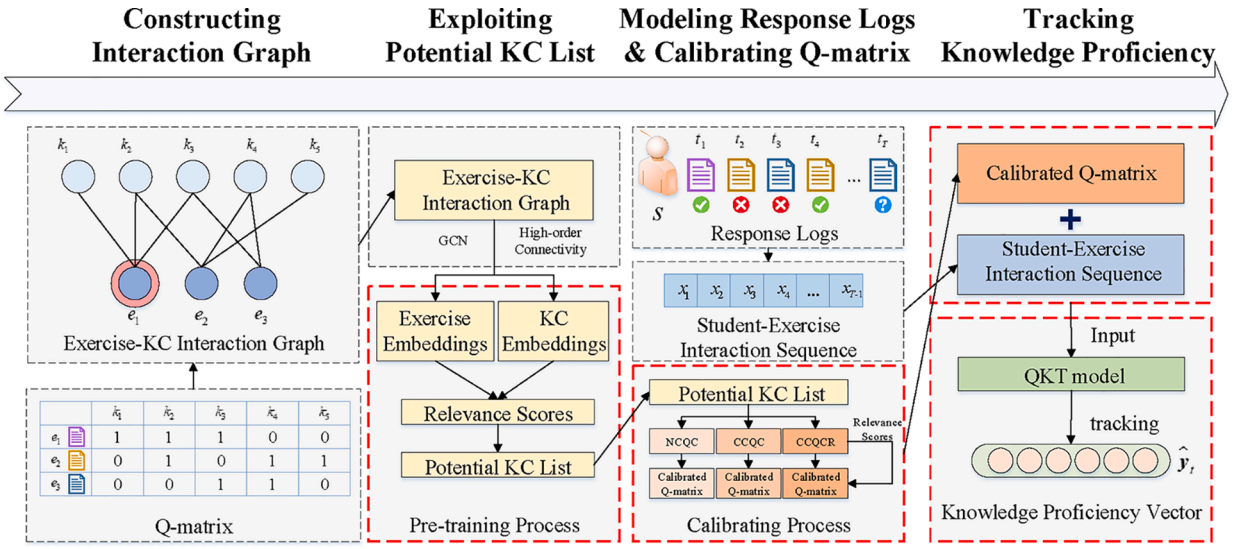


Fig. 2. Overview of the proposed CQKT framework.

compare some recent and improved knowledge tracing methods leveraging neural networks; lastly, the Q-matrix estimation methods are included, as the problem in question is related to the task of alleviating the subjective tendency in Q-matrix.

2.1. Cognitive diagnosis

Cognitive diagnosis is a critically promising direction in educational measurement, which has aroused considerable research interest due to its potential to provide diagnostic feedback about students' strengths and weaknesses (e.g. Ma & Jimmy, 2020a). Widely used cognitive diagnosis models (CDMs) could be divided into two aspects: saturated cognitive diagnosis models and constrained cognitive diagnosis models. Specifically, saturated cognitive diagnosis models denotes the G-DINA (generalized deterministic inputs, noisy "and" gate) model (Jimmy, 2011) as it is a generalized model which can be reduced to many constrained models by setting appropriate constraints.

Constrained cognitive diagnosis models, such as DINA (Jimmy, 2009), DINO (e.g. Templin & Henson, 2006) and A-CDM (Jimmy et al., 2011) can be obtained from the original G-DINA model under some certain constraints. For instance, DINA model is derived from the identity link G-DINA model by setting all parameters to zero except two (i.e. guessing and slipping). Also, depending on making some parameters of G-DINA be negative, DINO model can be directly attained. Similarly, without taking interaction effects into account, G-DINA model can be reduced to the A-CDM model.

Although effectively estimating students' KS by cognitive diagnosis models mentioned above has high educational impact, it is also inherently difficult to represent students' learning process by utilizing probabilistic models. Meanwhile, all those cognitive diagnosis models generally hypothesize that the KS of students on all certain KCs is binary (i.e. a KC is either in the 'mastered' or 'not mastered' state), which is not comprehensive enough to model such a dynamic changing learning process of students. Additionally, manually-labeled Q-matrix leveraged in those cognitive diagnosis models may be deficient because of inevitable errors and subjective bias (e.g. Liu, Xu, & Ying, 2012) which can negatively affect the estimation of the model parameters.

In this work, we focus on alleviating the problem of subjective tendency in Q-matrix and projecting students' KS on a certain KC into a continuous value in [0,1] to provide more fine-grained information about students' knowledge proficiency on each KC.

2.2. Knowledge tracing

Knowledge tracing (i.e. KT) task indirectly evaluates the students' knowledge proficiency of concepts on the basis of their response data. Recently many KT models based on deep learning show that there is a leap in score prediction performance of students. To the best of our knowledge, Deep Knowledge Tracing (DKT) is the first attempt to model students' complex learning process using neural network (e.g. Piech et al., 2015). Moreover, Dynamic Key-Value Memory Networks (DKVMN) (e.g. Zhang et al., 2017) introduces a Memory Augmented Neural Network (MANN) (e.g. Santoro, Bartunov, Botvinick, Wierstra, & Lillicrap, 2016) and is more interpretable than DKT to solve KT task using two matrices, *key* and *value*, which learns the correlation between exercises and the latent KCs and students' KS, respectively. Furthermore, some researches use graph convolutional network (GCN) related techniques for knowledge tracing, such as JKT (Song et al., 2021) and SEPN (Song et al., 2020). JKT can learn the embeddings of each exercise and its corresponding concepts through a two-layer graph convolutional network. SEPN is trained to learn the influence of students' demographic data and historical performance as well as their learning engagement to predict their performance. Since all these deep learning models enable to automatically discover underlying concepts of exercises, Q-matrix annotated by human experts is abandoned. Despite the success of these approaches, they do not guarantee to beat human experts in all datasets.

Another line of research includes methods via leveraging the rows of Q-matrix to represent exercises like Content-Based linear approach using Contextualized Q-matrix (CBCQ) and LSTM-Based contextual KT approach using Contextualized Q-matrix (LSTMCQ) (Huo et al., 2020b). Both of them tackle the KT problem depending on the contexts of learning, such as the difficulty of exercises, factors related to teaching, and connections with other KCs, etc., which are integrated into Q-matrix. Although these aforementioned methods take Q-matrix into consideration during the process of tracing students' KS and update Q-matrix by injecting contexts of learning into it, in essence, they fail to explore relationships between exercises and KCs and are not able to eliminate subjective tendency in Q-matrix.

In our method, we strive to model the relationships between exercises and KCs as an interaction graph so that we can exploit the high-order connectivity between exercises and KCs within the interaction graph using GCN in order to obtain a calibrated Q-matrix which can alleviate the issue of subjective bias in human-labeled Q-matrix.

2.3. Q-matrix estimation

Considering the subjective tendency of Q-matrix labeled by domain experts and possible misspecifications in Q-matrix, several Q-matrix estimation approaches have been developed to solve these problems. Basically, these approaches can be divided into two broad strands of works. One is the Q-matrix estimation method based on cognitive diagnosis model, and the other is depending on extra information (e.g. text information).

In effect, for the first kind of the Q-matrix estimation methods, they have been heavily studied in the area of educational measurement for many years and most of them have critically positive impact on calibrating Q-matrix. The G-DINA Discrimination Index (GDI) method is proposed in (e.g. Jimmy and Chiu, 2016), which can be used in conjunction with the G-DINA model. As an extension of GDI method, the stepwise method (e.g. Ma & Jimmy, 2020b) devised to validate Q-matrix for the sequential G-DINA model is based on GDI and Wald statistic. In literature (e.g. Wang et al., 2020b), three approaches are developed on the basis of the likelihood ratio test to estimate Q-matrix with a partial known Q-matrix and response data, which can be used with a wide variety of cognitive diagnosis models. The feasibility and effectiveness of the aforementioned methods have been demonstrated by extensive experiments, whereas all of them heavily rely on concrete cognitive diagnosis models. Furthermore, the process of estimation of Q-matrix is labour-intensive, with the premise of all parameters in corresponding cognitive diagnosis models already known.

On the other hand, recently, another kind of Q-matrix estimation methods based on extra information has been proposed and yielded great success. Specifically, in Neural Cognitive Diagnosis (NeuralCD) (e.g. Wang et al., 2020c) framework, a novel Q-matrix estimation method utilized a pre-trained Convolutional Neural Network (CNN) to predict KCs related to the text information of exercises is incorporated into the NeuralCD framework. Defining a partial order, this method connects the KCs estimated by CNN with those annotated by experts. Unfortunately, the partial order is not complete enough to cover every single KC so that we are not able to obtain the global relevance between all of KCs.

Thus, technically, our model select the second type of Q-matrix estimation method to calibrate Q-matrix. In particular, we further devise a more complete partial order over the whole KCs than NeuralCD and leverage GCN to generate those KCs which are more relevant to the input exercise as side information in order to eliminate the subjective tendency in Q-matrix.

3. Preliminaries

In this section, we introduce some preliminary knowledge about knowledge tracing, including necessary definitions and notations used in this paper and then present an overview for our solution.

3.1. Problem statement

Let $S = \{s_1, s_2, \dots, s_N\}$, $E = \{e_1, e_2, \dots, e_M\}$ and $Kn = \{k_1, k_2, \dots, k_K\}$, where N is the number of students, M is the number of exercises, and K is the number of knowledge concepts at a learning system, respectively. A student selects some exercises from E for practice, and his/her response logs $X = \{x_1, x_2, \dots, x_T\}$ are characterized by a set of tuple $x_t = (e_t, r_t)$, where $e_t \in E$ denotes the exercise attempted, $r_t \in \{0, 1\}$ denotes the correctness of the corresponding answer which is leaved by a student at timestep t. Besides, for explainable diagnosis of students' KS, a manually-labeled Q-matrix $\boldsymbol{Q} \in \mathbb{R}^{M \times N}$ which is a binary knowledge matrix is also required. If a KC k is involved in an exercise j, then $Q_{jk} = 1$; $Q_{jk} = 0$ otherwise. Without loss of generality, the research problem can be formulated as follows:

Definition. (*(Knowledge Tracing).*) *Given students' response logs R and the corresponding Q-matrix* $\boldsymbol{Q}$*, our goal is twofold: (1) predicting the knowledge proficiency on K KCs and response scores on next candidate exercise of each student at timestamp* $T + 1$*; (2) tracing the dynamics of KS of each student and estimating the level of how he/she maters on KCs from time 1 to T.*

3.2. Solution overview

Fig. 2 illustrates an overview of the proposed *Calibrated Q-matrix-based Knowledge Tracing* (CQKT) framework, consisting of four major steps: constructing interaction graph, exploiting potential KC list, modeling response logs & calibrating Q-matrix and tracking knowledge proficiency, respectively. The first step is devised to transform a raw Q-matrix into an exercise-KC interaction graph. Additionally, in the next step, the exercise embeddings and the KC embeddings are obtained from the interaction graph by using message passing mechanism based on GCN model and the high-order connectivity is exploited, as such, the potential KC list can be attained. Then, towards more realistic modeling, the response logs are required to be modeled as the student-exercise interaction sequences and the calibrated Q-matrix $\widehat{\boldsymbol{Q}}$can be estimated by three different methods (i.e. NCQC, CCQC and CCQCR) in the third step. Finally, with the help of the calibrated Q-matrix and the student-exercise interaction sequences, our framework has the mechanism to track the change of students' KS during their learning process and enables to predict their performance scores on exercise e_{t+1} in the last step. For better description, the key notations are summarized in Table 1.

4. Method

In this section, we first introduce the *Q-matrix-based Knowledge Tracing* (QKT) model, which is both a baseline and a premise of CQKT framework. Next, based on the QKT model, three Q-matrix calibration approaches based on a partial order are designed to eliminate the subjective tendency existed in Q-matrix. Last, we develop the CQKT framework which integrates the QKT model with the aforementioned Q-matrix calibration methods to conquer those problems defined in 3.1.

4.1. QKT

Following existing works (e.g. Huo et al., 2020a; Huo et al., 2020b), where relevant KCs are used for representing exercises, we develop a method called QKT which describes exercises by replacing all exercises' one-hot representation vectors with KC-based representation vectors. One reason for this shift of interest is that more explainable results of KS of students are what we exactly expect for.

Students' response logs $X = \{x_1, x_2, \dots, x_T\}$ are formulated as a set of tuple $x_t = (e_t, r_t)$. Herein, at time-step t, each row of Q-matrix as the raw exercise embedding $\boldsymbol{q}_t$ is a vector of a certain exercise in relation to one or more KCs:

Table 1
The key mathematical notations and descriptions.

Notation	Description
N	the total number of students
M	the total number of exercises
K	the total number of knowledge concepts
d	the embedding size
$\boldsymbol{Q}$	the human-labeled Q-matrix
$\widehat{\boldsymbol{Q}}$	the calibrated Q-matrix
$\boldsymbol{q}_t$	the exercise embedding based on related KCs at timestep t
$\widehat{y}_t$	the knowledge states/proficiency vector of a student at time step t
$\widehat{p}_t$	the probability of locating lender V at time t
rs	the relevance score between an exercise and a KC
RS_i	the relevance score set of exercise i with all KCs
V_i^n	the set of predicted top-n KCs of exercise e_i
D	the partial order relationship set

$$\boldsymbol{q}_t = \boldsymbol{e}_t^T \times \boldsymbol{Q} \tag{1}$$

where $\boldsymbol{e}_t$ indicates the one-hot representation vector of an exercise and the length of $\boldsymbol{e}_t$ is M.

To model such semantic rich and less dimensional exercise representations than DKT, we also take students' response logs into account, where the $\boldsymbol{x}_t = [\boldsymbol{q}_t, r_t]$ is fed into the QKT model and each of these vectors has a length of $K + 1$ as shown in Fig. 3. The output of QKT is a predicted vector of probabilities of getting each KC correct (i.e. the mastery level on each of KCs), denoted by $\widehat{y}_{t+1}$, and its length is equal to the number of KC. The inner product between the current predicted vector and the exercise vector represented by KCs in Q-matrix at next timestamp indicates the predicted score of the student's performance on e_{t+1}. During such a process, the flow of computation follows the set of procedures below:

$$\begin{aligned}
\boldsymbol{i}_t &= \sigma(\boldsymbol{W}_i \boldsymbol{x}_t + \boldsymbol{U}_i \boldsymbol{h}_{t-1} + \boldsymbol{b}_i) \\
\boldsymbol{f}_t &= \sigma(\boldsymbol{W}_f \boldsymbol{x}_t + \boldsymbol{U}_f \boldsymbol{h}_{t-1} + \boldsymbol{b}_f) \\
\boldsymbol{o}_t &= \sigma(\boldsymbol{W}_o \boldsymbol{x}_t + \boldsymbol{U}_o \boldsymbol{h}_{t-1} + \boldsymbol{b}_o) \\
\widehat{\boldsymbol{z}}_t &= tanh(\boldsymbol{W}\boldsymbol{h}_{t-1} + \boldsymbol{U}\boldsymbol{x}_t + \boldsymbol{b}) \\
\boldsymbol{z}_t &= \boldsymbol{f}_t \circ \boldsymbol{z}_{t-1} + \boldsymbol{i}_t \circ \widehat{\boldsymbol{z}}_t \\
\boldsymbol{h}_t &= \boldsymbol{o}_t \circ tanh(\boldsymbol{z}_{t-1}) \\
\widehat{\boldsymbol{y}}_t &= \sigma(\boldsymbol{h}_t)
\end{aligned} \tag{2}$$

where $\boldsymbol{i}_t$, $\boldsymbol{f}_t$ and $\boldsymbol{o}_t$ are the input, forget and output gates in the LSTM cell respectively. At timestamp t, $\widehat{z}_t$ is the memory cell candidate, z_t is the memory cell, $\boldsymbol{h}_t$ is the state of the hidden layer, $\widehat{y}_t$is the knowledge proficiency vector which elements indicate a student knowledge state on each KC, while $\boldsymbol{W}$, $\boldsymbol{W}_{i,\,f,\,o}$, $\boldsymbol{U}$, $\boldsymbol{U}_{i,\,f,\,o}$, $\boldsymbol{b}$ and $\boldsymbol{b}_{i,\,f,\,o}$, are the parameters to learn in the LSTM network. Moreover, $\circ$ is the element-wise product, $\sigma(\cdot)$ is the sigmoid function, and$tanh(\cdot)$is the hyperbolic tangent non-linear activation function.

After the probability of getting each KC correctly has been computed, the corresponding exercise vector in the Q-matrix can be leveraged to compute the predicted value$\widehat{p}_t$. As the output layer adopts a sigmoid activation function on top of the linear transformation, the value of each element in $\widehat{y}_t$ is independent of the others:

$$\widehat{p}_t = \widehat{\boldsymbol{y}}_t^T \times \boldsymbol{q}_{t+1} \tag{3}$$

The loss function of QKT is the cross entropy between the output $\widehat{p}_t$ and the true label r_{t+1}:

$$loss_{QKT} = -\frac{1}{T}\sum_{t=0}^{T-1}[r_{t+1} log(\widehat{p}_t) + (1 - r_{t+1}) log(1 - \widehat{p}_t)] \tag{4}$$

After training, the value of $\widehat{y}_t$ is the diagnosis result, which denotes the student's knowledge proficiency on each KC at timestep t.

4.2. Q-matrix calibration

Considering that a KC is usually associated with several exercises and an exercise can also cover many KCs, it is naturally to represent them as a bipartite interaction graph, where vertices are KCs and exercises. Notably, edges which exist between exercises and KCs in the exercise-KC interaction graph have been proved to present semantic signals (e.g. Liu, Yang, & Chen, 2020). This insight motivates us to exploit high-order connectivity (e.g. Wang, He, Wang, Feng, & Chua, 2019) from exercise-KC interactions, a straightforward way that encodes collaborative signal in the interaction graph structure to refine Q-matrix.

4.2.1. Relevance score computation

Fig. 4 illustrates the concept of high-order connectivity. The exercise with probable bias for calibration is e_1, labeled with double circle in the left area of exercise-KC interaction graph. The right area presents three structure that is expanded from e_1. The high-order connectivity indicates the path that reaches e_1 from any node with the path length l larger than 1. There are rich semantic information embodied in such high-order connectivity and which carries collaborative signal. For instance, the path $e_1 \leftarrow k_2 \leftarrow e_2$indicates the content similarity, as both of two exercises involves the same KC; the longer path $e_1 \leftarrow k_2 \leftarrow e_2 \leftarrow k_4$suggests that e_1 is likely to cover k_4, since its neighbor exercise e_2 contains k_4. Additionally, from the holistic perspective of $l = 3$, k_4 is more likely to be the candidate knowledge point which is going to be used for calibrating Q-matrix as side information than k_5, as there are two paths connecting $< k_4, e_1 >$, while only one path connects $< k_5, e_1 >$.

Inspired by high-order connectivity, we pre-train a GCN model (e.g. He et al., 2020) to obtain the embeddings of all vertices in the exercise-KC interaction graph for predicting relevance scores between the input exercise and KCs, where the human-labeled Q-matrix is utilized as label for training. GCN has the advantage of extracting semantic signals from the exercise-KC interaction graph, thus it is able to capture potential KCs for the input exercise. Specifically, first, following mainstream GCN models, we describe an exercise e_i (a knowledge concept k_j) with an embedding vector $\boldsymbol{e}_{e_i} \in \mathbb{R}^d (\boldsymbol{e}_{k_j} \in \mathbb{R}^d)$, where d denotes the embedding size. Then, the pre-trained network takes the embedding vector $\boldsymbol{e}_{e_i}$ and the embeddings of all KCs as input, and output the relevance score set RS_i of those KCs related to e_i:

$$rs_{e_i,k_j} = sigmoid(\boldsymbol{e}_{e_i}^T \cdot \boldsymbol{e}_{k_j}) \tag{5}$$

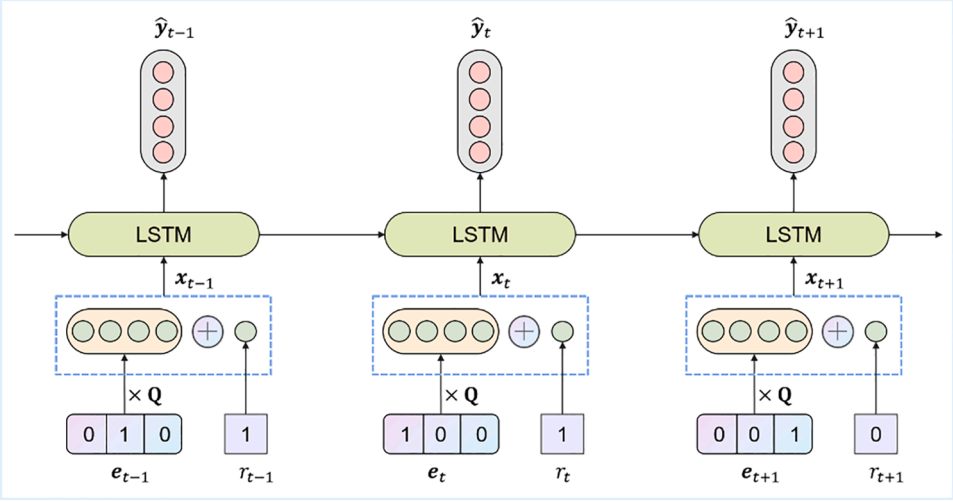


Fig. 3. The framework of the QKT model.

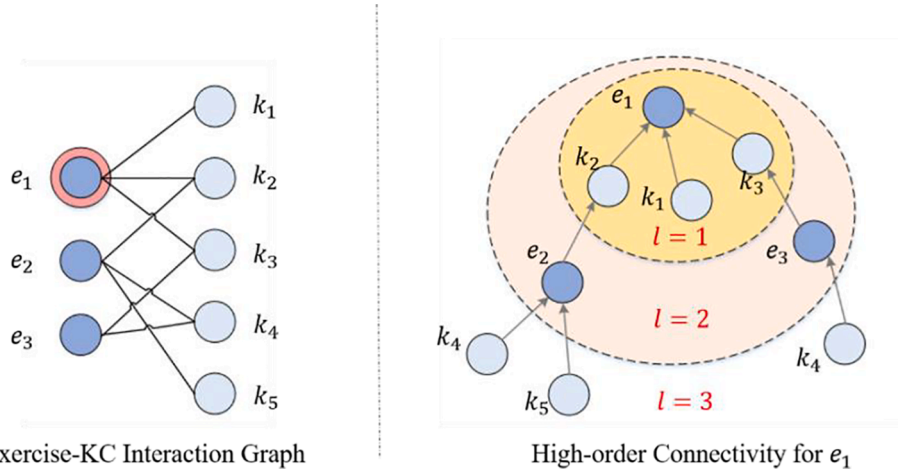


Fig. 4. An illustration of the exercise-KC interaction graph and high-order connectivity. The node e_1 is the target exercise to provide calibrations for.

$$RS_i = \left\{rs_{e_i,k_j}\right\}_{j=1}^{K} \tag{6}$$

After descending sort all relevance scores in RS_i, we define $V_i^n = \{V_{ij_1}, V_{ij_2}, \ldots, V_{ij_n}\}$ as the set of top-n KCs of exercise e_i according to RS_i. Thus we develop three methods to calibrate Q-matrix by combining V_i^n with Q-matrix: a naive confidence Q-matrix calibration method (**NCQC**), a complete confidence Q-matrix calibration method (**CCQC**) and a complete confidence Q-matrix calibration plus relevance (**CCQCR**) respectively.

4.2.2. NCQC

In view of the aforementioned idea that KCs annotated by domain experts have higher confidence than those predicted by the pre-trained model, we consider that KCs labeled by Q-matrix are more relevant to e_i than concepts in $\left\{k_j | k_j \in V_i^n \wedge Q_{ij} = 0\right\}$. To accomplish this, we adopt a pairwise Bayesian treatment as follows. For convenience, we define a partial order $>_i^+$ as:

$$a >_i^+ b \tag{7}$$

where $a >_i^+ b$ suggests that the KC a is more important than the KC b in exercise e_i and $Q_{ia} = 1 \wedge b \in \left\{k_j | k_j \in V_i^n \wedge Q_{ij} = 0\right\}$. Then we define the partial order relationship set as:

$$D_{NCQC} = \left\{(i, a, b) | a >_i^+ b, i = 1, 2, ..., M\right\} \tag{8}$$

Following traditional Bayesian treatment, we assume $\widehat{\boldsymbol{Q}} \in \mathbb{R}^{M \times K}$ follows a zero mean Gaussian prior with standard deviation σ of each dimension. In order to assign higher confidence to Q-matrix labels, we define $p(a >_i^+ b | \widehat{\boldsymbol{Q}}_i)$ with a pairwise logistic-like function:

$$p(a >_i^+ b | \widehat{\boldsymbol{Q}}_i) = \frac{1}{1 + e^{-\lambda(\widehat{Q}_{ia} - \widehat{Q}_{ib})}} \tag{9}$$

The hyperparameter λ controls the discrimination of relevance values between labeled and unlabeled KCs. Consequently, the log posterior probability over D_{NCQC} on $\widehat{\boldsymbol{Q}}$ is finally formulated as:

$$\begin{aligned}
ln p(\widehat{\boldsymbol{Q}} | D_{NCQC}) &= ln \prod_{(i,a,b) \in D_{NCQC}} p(a >_i^+ b | \widehat{\boldsymbol{Q}}_i) p(\widehat{\boldsymbol{Q}}_i) \\
&= \sum_{(i,a,b) \in D_{NCQC}} ln p(a >_i^+ b | \widehat{\boldsymbol{Q}}_i) + \sum_{i=1}^{M} p(\widehat{\boldsymbol{Q}}_i) \\
&= \sum_{i=1}^{M} \sum_{a=1}^{K} \sum_{b=1}^{K} I(a >_i^+ b) ln \frac{1}{1 + e^{-\lambda(\widehat{Q}_{ia} - \widehat{Q}_{ib})}} + \sum_{i=1}^{M} \sum_{j=1}^{K} ln \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{\widehat{Q}_{ij}^2}{2\sigma^2}} \\
&= \sum_{i=1}^{M} \sum_{a=1}^{K} \sum_{b=1}^{K} I(a >_i^+ b) ln \frac{1}{1 + e^{-\lambda(\widehat{Q}_{ia} - \widehat{Q}_{ib})}} + \sum_{i=1}^{M} \sum_{j=1}^{K} ln \frac{1}{\sqrt{2\pi}\sigma} - \sum_{i=1}^{M} \sum_{j=1}^{K} \frac{\widehat{Q}_{ij}^2}{2\sigma^2} \\
&= \sum_{i=1}^{M} \sum_{a=1}^{K} \sum_{b=1}^{K} I(a >_i^+ b) ln \frac{1}{1 + e^{-\lambda(\widehat{Q}_{ia} - \widehat{Q}_{ib})}} + C - \sum_{i=1}^{M} \sum_{j=1}^{K} \frac{\widehat{Q}_{ij}^2}{2\sigma^2}
\end{aligned} \tag{10}$$

where C is a constant which can be ignored during optimization and $I(\cdot)$ is an indicator that equals to 1 if the condition '$\cdot$' is satisfied. Ultimately, the calibrated Q-matrix $\widehat{\boldsymbol{Q}}$ is naively estimated by NCQC approach without completely covering partial order relationships between all KCs.

4.2.3. CCQC

Based on the NCQC method, CCQC method further distills partial order relationships between KCs with the pre-trained model. For this, we define a more complete partial order as:

$$a_{expert\&model} >_i^+ b_{expert\&\overline{model}} >_i^+ c_{\overline{expert}\&model} >_i^+ d_{\overline{expert}\&\overline{model}} \tag{11}$$

where a, b, c, and d are the elements of knowledge concept set Kn. Here $a_{expert\&model}$ denotes the KC a which is both annotated by expert and can also be found in V_i^n, while $b_{expert\&\overline{model}}$ implies that KC b is labeled by domain expert in Q-matrix but does not exist in V_i^n. Analogous to the KC a and the KC b, the meanings of $c_{\overline{expert}\&model}$ and $d_{\overline{expert}\&\overline{model}}$ can be interpreted in a similar way. Along this line, we are able to draw a conclusion that 'a is more important than b in e_i'. Thereby, according to Eq. (11), we define the partial order relationship set as:

$$D_{CCQC} = \left\{(i, k_r, k_s) | k_r >_i^+ k_s, i = 1, 2, ..., M\right\} \tag{12}$$

Analogous to *NCQC*, the log posterior probability over D_{CCQC} on $\widehat{\boldsymbol{Q}}$ is finally formulated as:

$$lnp(\widehat{\boldsymbol{Q}}|D_{CCQC}) = ln \prod_{(i,k_r,k_s)\in D_{CCQC}} p(k_r >_i^+ k_s|\widehat{\boldsymbol{Q}}_i)p(\widehat{\boldsymbol{Q}}_i)$$
$$= \sum_{i=1}^{M}\sum_{k_r=1}^{K}\sum_{k_s=1}^{K} I(k_r >_i^+ k_s) ln \frac{1}{1+e^{-\lambda(\widehat{Q}_{ik_r}-\widehat{Q}_{ik_s})}} + C - \sum_{i=1}^{M}\sum_{j=1}^{K}\frac{\widehat{Q}_{ij}^2}{2\sigma^2} \tag{13}$$

By maximizing $lnp(\widehat{\boldsymbol{Q}}|D_{CCQC})$ until it is converged, the calibrated Q-matrix $\widehat{\boldsymbol{Q}}$ can be attained.

4.2.4. CCQCR

Although CCQC generally outperforms NCQC owing to its comprehensiveness and completeness, yet it still ignores differences between KCs obtained by pre-trained model. For example, there are two KCs $a^{(g)}_{expert\&model}$ and $a^{(h)}_{expert\&model}$which are both labeled by expert and are predicted by the pre-trained model. In this situation, we can only argue that both of them are significant in an exercise while have no idea that which one is more significant. To this end, we propose a partial order with relevance score, that is:

$$\underbrace{a^{(g)}_{expert\&model\&(rs_{e_i,k_m}>rs_{e_i,k_n})} >_i^+ a^{(h)}_{expert\&model\&(rs_{e_i,k_m}>rs_{e_i,k_n})}}_{a_{expert\&model}} >_i^+$$
$$b_{expert\&\overline{model}} >_i^+$$
$$\underbrace{c^{(p)}_{\overline{expert}\&model\&(rs_{e_i,k_p}>rs_{e_i,k_q})} >_i^+ c^{(q)}_{\overline{expert}\&model\&(rs_{e_i,k_p}>rs_{e_i,k_q})}}_{c_{\overline{expert}\&model}} >_i^+$$
$$d_{\overline{expert}\&\overline{model}} \tag{14}$$

where the KC $a^{(g)}_{expert\&model}$ is more significant than $a^{(h)}_{expert\&model}$ in e_i due to $rs_{e_i,k_g} > rs_{e_i,k_h}$even if both of them are annotated by expert and are obtained by model. With the help of Eq. (14), the partial order relationship set can be formulated as:

$$D_{CCQCR} = \{(i,k_r,k_s)|k_r >_i^+ k_s, i = 1,2,...,M\} \tag{15}$$

In this way, the log posterior probability over D_{CCQCR} on $\widehat{\boldsymbol{Q}}$ is finally defined as:

$$lnp(\widehat{\boldsymbol{Q}}|D_{CCQCR}) = ln \prod_{(i,k_r,k_s)\in D_{CCQCR}} p(k_r >_i^+ k_s|\widehat{\boldsymbol{Q}}_i)p(\widehat{\boldsymbol{Q}}_i)$$
$$= \sum_{i=1}^{M}\sum_{k_r=1}^{K}\sum_{k_s=1}^{K} I(k_r >_i^+ k_s) ln \frac{1}{1+e^{-\lambda(\widehat{Q}_{ik_r}-\widehat{Q}_{ik_s})}} + C - \sum_{i=1}^{M}\sum_{j=1}^{K}\frac{\widehat{Q}_{ij}^2}{2\sigma^2} \tag{16}$$

Again, we need to maximize $lnp(\widehat{\boldsymbol{Q}}|D_{CCQCR})$ to calibrate Q-matrix for alleviating the issue of subjective bias within it. For better comparison, we present the differences between three methods in Fig. 5.

4.3. CQKT+

Before using $\widehat{\boldsymbol{Q}}$ to improve QKT, we need to restrict its elements to the range [0, 1], and let the elements of KCs which are unlabeled or are not predicted in top-n KC list be 0. Therefore, $sigmoid(\widehat{\boldsymbol{Q}})^{\circ}\boldsymbol{M}$ is leveraged to be the substitution of Q-matrix in QKT, where $\boldsymbol{M} \in \{0,1\}^{M\times K}$ is a mask matrix, and $M_{ij} = 1$ if $j \in V_i^n$ or $Q_{ij} = 1$; $M_{ij} = 0$ otherwise. To track knowledge proficiency of students and calibrate Q-matrix simultaneously, we combine all the loss functions to form a joint optimization framework, namely, we solve:

$$minloss_{CQKT+} = min[-lnp(\widehat{\boldsymbol{Q}}|D_*) + \sum_{i=1}^{N} loss_i^{QKT}] \tag{17}$$

where $* \in \{NCQC, CCQC, CCQCR\}$ and $\widehat{\boldsymbol{Q}}$ is trained together with the QKT model.

5. Experiments

In this section, we conduct extensive experiments to evaluate the performance of our proposed CQKT framework. Specifically, we describe the datasets and introduce the experimental setup and we also demonstrate the effectiveness of our framework on three educational tasks including score prediction, knowledge state estimation, and diagnosis result visualization. The details of the experiments will be discussed below.

5.1. Datasets

We evaluate performance of our model and several baseline models on predicting students' response and estimating students' knowledge proficiency using three real-world KT datasets. Table 2 summarizes basic statistics of the datasets.

5.1.1. ASSISTment09

The first dataset is *ASSISTment09*[1] (ASSISTments2009-2010 skill builder), an open dataset collected by the ASSISTments online tutoring systems, which is widely used by researchers for purposes such as KT and student modeling. We choose the public corrected version which mitigates the duplicated data problem proposed by previous research (e.g. Wang et al., 2020c), which filter out students with less than 15 response logs. Notably, in ASSISTment09, 3 KCs were attempted 6 times or less, so they are also removed as outliers for inadequate sequence length, leaving 120 KCs in total.

5.1.2. Intellilence18

The second dataset is *Intellilence18* supplied privately by Oneoftops Ltd., a company offering teaching and learning resources to middle and high school teachers and students, which provides language learning and practicing services to young language learners (e.g. Huo et al., 2020b). This dataset contains records of 1452 students involving 159 KCs, and a total of 256,612 interactions. On average, students made 177 attempts and 1616 attempts were observed for each KC. Moreover, we also filter out students with less than 30 response logs.

5.1.3. KDD2010

The third dataset is Bridge Algebra 2006, which stems from the KDD Cup 2010 EDM Challenge[2] (Zhang, Zhu, Zhang, Ji, Pan, & Yin, 2021). There are 1145 students, 129,255 questions and 493 concepts. We get in total 1,817,016 observed responses. For this dataset, we considered the combination of question and the step IDs as one answer which is recommended by challenge organizers. The largest number of concepts related with a question is 5. We also excluded concept tags that are unnamed or dummies to remove the noise for three datasets and only use the sequence of students whose history records are longer than 3

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009–2010-data.
[2] https://pslcdatashop.web.cmu.edu/KDDCup/downloads.jsp

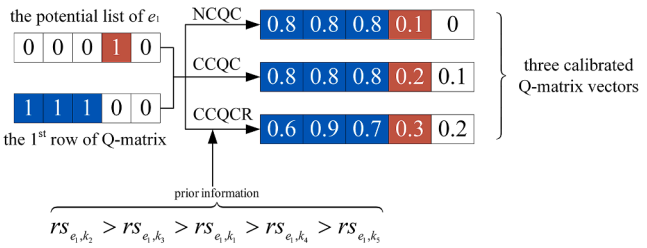


Fig. 5. The differences between three Q-matrix calibration methods.

Table 2
Dataset summary.

Summary	Dataset		
	ASSISTment09	*Intellilence18*	*Bridge Algebra2006*
No. of KCs	120	159	493
No. of attempts	337,990	256,612	1,817,016
No. of students	4,163	1,452	1,145
No. of exercises	17,751	23,373	129,255
Avg. attempts per student	81	177	169
Avg. attempts per KC	2,817	1,616	1,708

because history exercise records that are too short are meaningless to predict.

5.2. Experimental setup

We set hyperparameters $\lambda = 0.1$ (Eq. (9)) and $\sigma = 1$ (Eq. (10)). For n in top-n KCs filtering, the value we utilize can make the predicting GCN model reach 0.85 recall. Thereby, in our experiment, $n = 20$. We implement the initialization of the parameters depending on *Xavier* (e.g. Glorot & Bengio, 2010), which fill the weights with random values sampled from $\mathcal{N}(0, std^2)$, where $std = \sqrt{2/(n_{in} + n_{out})}$, n_{in} is the number of neurons feeding into the weights, n_{out} is the number of neurons which the results are sent to.

In CQKT framework (i.e. CQKT + models), we leverage 3 layers to capture high-order connectivity. Meanwhile, we pre-train the model with Adam (e.g. Kingma & Ba, 2015) and use the default learning rate of 0.001 and default mini-batch size of 512. Bayesian Personalized Ranking (BPR) loss is used as loss function for pre-training the model.

To evaluate the performance of our CQKT + models, we compare them with previous approaches. The details of them are as follows:

(1) *DKT*: DKT[3] is an implementation using a LSTM which leverages a series of one-hot representations of exercise and trains a unified network for predicting students' next interaction on certain exercise (e.g. Piech et al., 2015).
(2) *DKVMN*: DKVMN[4] uses a key-value memory network to store the KCs' underlying concept representations and states (e.g. Zhang et al., 2017).
(3) *LSTMCQ*: LSTMCQ is implemented by an RNN with LSTM cells. The architecture of it is similar to that of DKT, but the one-hot input vectors are replaced with the contextualized representation (e.g. Huo et al., 2020b).
(4) *QKT*: This is our extension to the DKT model, which directly uses KCs pertaining to exercise in conjunction with students' response log as the input of DKT and predicts students' response for each question.
(5) *CQKT + NCQC*: This is one of our proposed knowledge tracing methods, using the calibrated Q-matrix with a naive confidence Q-matrix calibration method.
(6) *CQKT + CCQC*: This is our implementation of CQKT with a complete confidence Q-matrix calibration method.
(7) *CQKT + CCQCR*: This is also one of our proposed methods. It adds relevance score between exercise and KC as prior information into the pre-defined partial order to accurately calibrate Q-matrix for knowledge tracing.

We perform 80%/20% train/test split of each student's historical interactions and all the models are evaluated with 5-fold cross validation.

5.3. Experimental results on three tasks

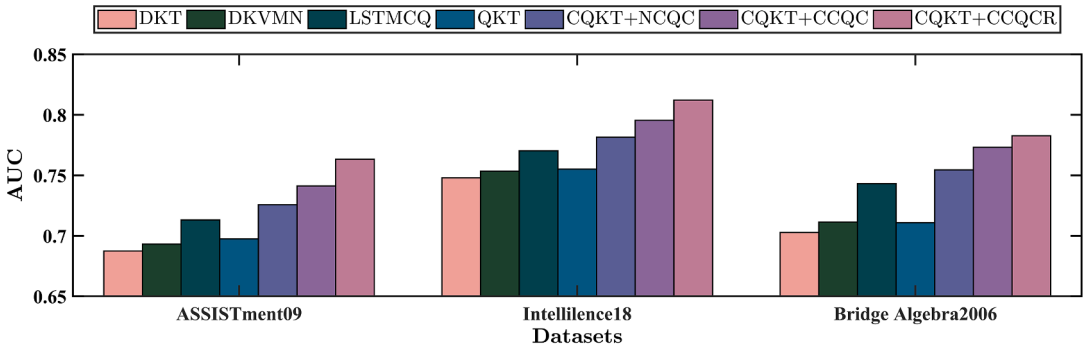
In this subsection, we compare the performance of all the baselines with our proposed framework on three educational tasks to demonstrate the effectiveness and the interpretation of the CQKT + models on the basis of CQKT framework.

5.3.1. Student score prediction

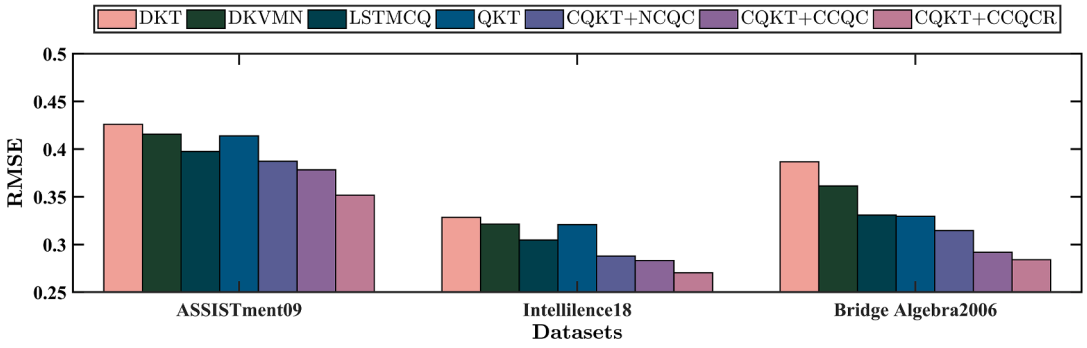
The first educational task is to predict student performances on exercises in the future, where we assess the predictive performance of our models. With trained CQKT + models, we leverage these models to predict whether or not a student can correctly answer a certain exercise at time $T + 1$. In this experiment, we select all the baselines aforementioned in Section 5.2 for comparison. Moreover, we select evaluation metrics from both classification perspective and regression perspective, including the Area Under the Curve of the ROC (AUC, the higher the better) and the Root Mean Square Error (RMSE, the lower the better) (e.g. Su, Liu, Liu, Huang, Yin, Chen, Ding, Wei, & Hu, 2018).

Fig. 6 presents the holistic results of all models for predicting scores of students. To be specific, there are several observations from the figure: First, the methods adopting Q-matrix (i.e., LSTMCQ, QKT, CQKT +) outperform almost the ones without Q-matrix. This demonstrates that it is effective to take association between exercises and KCs with Q-matrix into modeling for prediction. However, there is an exception in Fig. 6, where DKVMN performs comparably with QKT. This has two possible implications: (1) the utility of MANN in DKVMN helps to offset the insufficiency of a non-linear model using embarrassingly simple one-hot exercise representation without taking Q-matrix into consideration (e.g. Zhang et al., 2017); (2) as the QKT results do not support the former statement, it is more probable that the deficiency in the QKT is caused by the subjective bias in Q-matrix. This can be further validated in Table 3. There, in *ASSISTment09*, the improvement in the AUC from QKT to CQKT + NCQC is 0.0264, whereas it is only 0.0072 from DKT to QKT. The improvement brought about by eliminating subjective bias in Q-matrix is 3.6 times as much as that brought about by simply using KC as exercise representation. In effect, CQKT + NCQC shows better performance than QKT in general, by, on average, 0.0201 in AUC and 0.0247 in RMSE. Both are directly significant improvements. This is the strongest evidence of effectiveness caused by Q-matrix calibration. Second, among those methods using Q-matrix, the methods with Q-matrix calibration consistently outperform the ones without (i.e., LSTMCQ, QKT), which indicates the effectiveness of the strategies proposed in Section 4.2 and the rationality of exploiting the high-order connectivity between exercise and KCs. Third, an interesting finding is that, among those methods using calibrated Q-matrix, CQKT + CCQCR performs better than CQKT + CCQC and CQKT + CCQC performs better than NCQC, which shows the limitation of incomplete partial order between KCs and the significance of leveraging prior information (i.e., relevance score) obtained by pre-trained model as a booster. In summary, all these evidences demonstrate the effectiveness and rationality of our proposed models.

[3] https://github.com/mmkhajah/dkt
[4] https://github.com/jennyzhang0215/DKVMN



(a) AUC results of different models.



(b) RMSE results of different models.

Fig. 6. AUC and RMSE of the compared methods on three datasets.

Table 3
Comparison of methods: AUC for classification and RMSE for regression.

Model	*ASSISTment09*		*Intellilence18*		*Bridge Algebra2006*	
	AUC	RMSE	AUC	RMSE	AUC	RMSE
DKT	0.6874	0.4260	0.7479	0.3285	0.7027	0.3867
DKVMN	0.6931	0.4156	0.7534	0.3214	0.7113	0.3613
LSTMCQ	0.7131	0.3975	0.7703	0.3047	0.7431	0.3308
QKT	**0.6974**	**0.4138**	**0.7551**	**0.3208**	**0.7108**	**0.3296**
CQKT + NCQC	0.7257	0.3872	0.7815	0.2878	0.7545	0.3146
CQKT + CCQC	0.7412	0.3783	0.7954	0.2831	0.7731	0.2918
CQKT + CCQCR	**0.7632**	**0.3517**	**0.8122**	**0.2704**	**0.7826**	**0.2841**

5.3.2. Knowledge proficiency estimation

The second crucial ability of our framework is to estimate the knowledge proficiency of each student in the future. In order to evaluate the effectiveness of our framework, i.e., whether or not the estimation of KS at next time $T+1$ is elegant, we conduct the following experiment.

Actually, it is inherently difficult to directly evaluate the knowledge states of student, since there is no general approach to obtain the actual value of student's proficiency levels on KCs. As an alternative, following the reference (e.g. Huang et al., 2020), we adopt a ranking-based evaluation experiment. Intuitively, if student a masters better on a specific KC k than student b at time $T+1$, then student a will be more likely to answer the exercises related to concept k correctly than student b at that time (e.g. Chen et al., 2017). Thus, we utilize Degree of Agreement (DOA) (e.g. Fouss, Pirotte, Renders, & Saerens, 2007) as the evaluation metric of this type of ranking performance. For a specific KC k, $DOA(k)$ is defined as:

$$DOA(k) = \frac{1}{Z}\sum_{a=1}^{N}\sum_{b=1}^{N} I(KS_{ak} > KS_{bk}) \sum_{j=1}^{M} q_{jk}\left[I(r_{aj}=1) \wedge I(r_{bj}=0)\right] \tag{18}$$

where $Z = \sum_{a=1}^{N}\sum_{b=1}^{N} I(KS_{ak} > KS_{bk}) \sum_{j=1}^{M} q_{jk}$, KS_{ak} is the proficiency of student a on KC k. $I(\cdot)$is an indicator that equals to 1 if condition '$\cdot$' is satisfied otherwise equals to 0. q_{jk} indicates the corresponding element of the raw Q-matrix or the calibrated Q-matrix and r_{aj} denotes the response log of student b on exercise j. We average $DOA(k)$ on all KCs to evaluate the quality of diagnostic result (i.e., knowledge proficiency

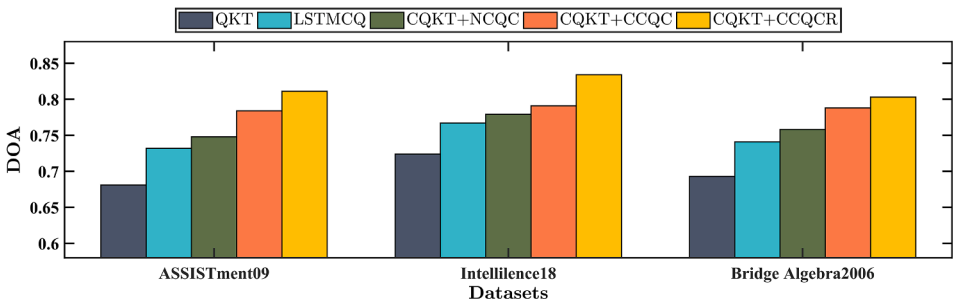


Fig. 7. DOA results of different models on three datasets.

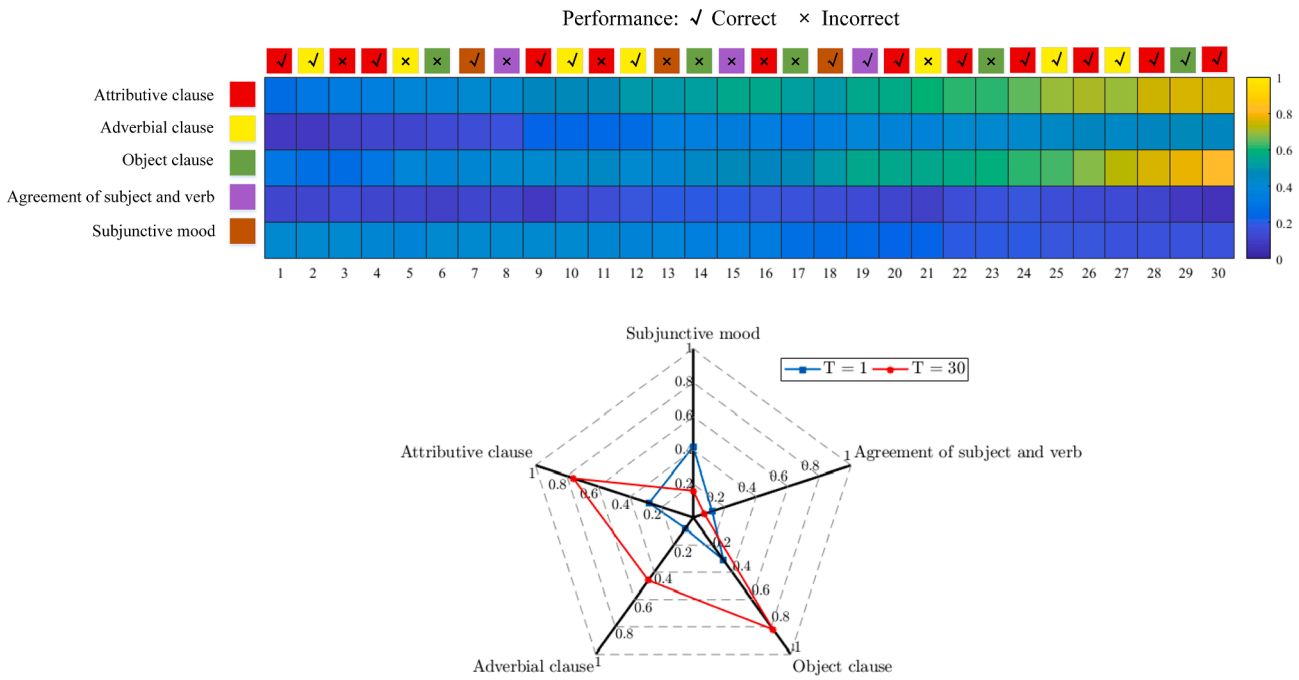


Fig. 8. An instance of the knowledge mastery level tracking of a certain student on five concepts during his 30 exercising steps, which is painted in the top heatmap. Left side presents all concepts, which are labeled in different colours. Top line indicates his performance on the 30 exercises. Bottom radar figure exhibits his knowledge mastery levels on five KCs after practicing from T = 1 to T = 30.

attained by models).

For model comparisons, we choose LSTMCQ and QKT as baselines, since all other models (i.e., DKT, DKVMN) are unexplainable for the diagnosis. Fig. 7 presents the experimental results. From the figure we can observe that DOAs of CQKT+ (i.e., CQKT + NCQC, CQKT + CCQC and CQKT + CCQCR) are significantly higher than baselines, which proves that knowledge proficiencies estimated by them are reasonable. The DOAs of CQKT + CCQCR and CQKT + CCQC are much higher than CQKT + NCQC, which indicates that a more complete partial order can contribute to estimating knowledge proficiency of students accurately. DOA of CQKT + CCQCR is slightly higher than CQKT + CCQC due to the use of prior information. Besides, the DOA of LSTMCQ outperforms QKT because of the contextualization of Q-matrix in LSTMCQ.

5.3.3. *Diagnosis result visualization*

As mentioned before, our proposed CQKT framework performs well on tracing the knowledge proficiency of a specific student in an explainable manner on the basis of the student's proficiency vectors $\hat{y}$ at different timestamps.

Fig. 8 provides a case study of visualizing the diagnosis results of a student on five knowledge concepts in Intellilence18 (we only show five KCs for better illustration). From the figure, he continuously makes progress on Attributive Clause from T = 1 (0.28) to T = 30 (0.76), since he practices the increasing number (e.g., 1, 3, 4, et al.) of exercises related to Attributive Clause during this period. However, his mastery levels on Subjunctive Mood declines gradually following the heatmap and the radar figure (from 0.42 to 0.16). We notice that he only attempts very few relevant exercises (i.e., 7, 13, 18) at each time, and thus he may forget the Subjunctive Mood knowledge. These observations imply that he needs a timely review on Subjunctive Mood. Based on the diagnostic result visualization, the framework has the ability to provide more personalized training services for him in practice.

6. Conclusions and future work

In this paper, we provided a focused study on dynamically diagnosing the knowledge proficiency of students. We designed a *Calibrated Q-matrix-based Knowledge Tracing* (CQKT) framework for KT task and proposed three Q-matrix calibration approaches to address the problem of subjective bias in Q-matrix labeled by domain experts. Specifically, we first attempted to exploit the high-order connectivity between exercises and KCs to obtain potential KCs of each exercise by using graph convolution network. Then, three pairwise Bayesian Q-matrix calibration approaches were devised to refine and calibrate the raw Q-matrix to eliminate the subjective tendency of the Q-matrix defined by subject matter experts. After that, the embedding of each exercise aggregated the calibrated Q-matrix with historical student interactions was fed into the Long Short-Term Memory (LSTM) network to tracking students' knowledge states. The results of extensive experiments conducted on three real-world benchmark datasets confirmed that our method could trace the knowledge proficiency of students effectively and we demonstrated the practicability of this method depending on applying it to three fundamental diagnostic tasks, including score prediction, knowledge state estimation, and diagnosis result visualization. However, two main limitations still exist in our framework. On one hand, three Q-matrix calibration methods are deeply entangled with the knowledge tracing model and fail to generalize on other KT model architectures. On the other hand, we solely exploit high-order connectivity between exercises and KCs, ignoring underlying semantic information contained in the student-exercise-KC interaction. In future, we will pay attention to making the model more flexible and mine latent semantic information contained in the student-exercise-KC interaction. Also, depending on the estimated knowledge states of students, we can make recommendation of exercises to students towards their weakness, which can be easily explained and provide meaningful feedbacks to students.

CRediT authorship contribution statement

Wentao Wang: Conceptualization, Methodology, Software. **Huifang Ma:** Supervision, Writing – review & editing, Funding acquisition. **Yan Zhao:** Visualization, Investigation. **Zhixin Li:** Supervision, Funding acquisition. **Xiangchun He:** Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: We would like to submit the manuscript entitled' Tracking Knowledge Proficiency of Students with Calibrated Q-Matrix', which we wish to be considered for publication in Expert Systems with Applications. No conflict of interest exits in the submission of this manuscript, and manuscript is approved by all authors for publication.

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